MIT | Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	COURSE SYLLABUS	
Department of Computer Science & Engineering(AIML)	**W.E.F**	**AY:** 2025 - 2026 (Rev. 2023 (NEP))
SECOND YEAR BACHELOR OF TECHNOLOGY CSE(AIML)	**COURSE NAME**	APPLIED MATHEMATICS
	COURSE CODE	2301257T
	COURSE CREDITS	3
RELEASE DATE : 01/07/2025	**REVISION NO.**	0.0

TEACHING SCHEME (HOURS/WEEK)		**EXAMINATION SCHEME AND MARKS**					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	TUTO-RIAL/PRACTICAL	
3	NIL	30	20	50	NIL	NIL	100

PREREQUISITE KNOWLEDGE: NIL

COURSE OBJECTIVES:

2301257T.CEO.1: To evaluate matrix properties and solve linear systems using vector space concepts.
2301257T.CEO.2: To apply probability distributions and hypothesis testing for statistical problem solving.
2301257T.CEO.3: To apply Markov analysis to model real world stochastic systems.
2301257T.CEO.4: To solve theoretically the problems of regression and numerical method and calculus applications.

COURSE OUTCOMES:

After successful completion of the course, students will be able to,

2301257T.CO. 1: Compute the eigen values and eigen vectors of a matrix, apply vector operations. [L3]
2301257T.CO. 2: Analyze real world problems using probability distribution and hypothesis testing. [L4]
2301257T.CO. 3: Model sequential data using Markov chains. [L3]
2301257T.CO. 4: Solve regression,numerical method and calculus applications. [L3]

COURSE ABSTRACT:

This course is designed to build a strong mathematical foundation for students pursuing engineering. It introduces essential topics such as Linear Algebra, Probability and Statistics, Hypothesis Testing, Stochastic Processes, Regression Analysis, and Numerical Methods. Students will learn to model, analyze, and interpret data-driven problems through theoretical and applied mathematical techniques. The course emphasizes structured problem-solving, logical reasoning, and critical thinking, helping students to build a solid conceptual understanding applicable across multiple engineering domains. By mastering these mathematical tools, learners will be better equipped to analyze data, identify patterns, and solve complex real-world problems with precision and clarity.

THEORY COURSE CONTENTS

UNIT 1	LINEAR ALGEBRA	12 HOURS

Application/Case Study: Text Data Representation using Vectors (Bag of Words Model)
Contents: Basic Concepts, Matrix Addition, Scalar Multiplication, Matrix Multiplication, Linear Systems of Equations, Rank of a Matrix, Eigenvalues, Eigenvectors, Symmetric Matrices, Skew-Symmetric Matrices, Orthogonal Matrices. Similarity of Matrices, Diagonalization, Vector Spaces, Basis, Dimension of vector spaces, Linear Independence, Inner Product Spaces. Vector Operations: Addition/subtraction, scalar multiplication, Dot product (projections, similarity), Cross product (3D applications), Vector Calculus: partial and first-order derivatives of vectors.
Self Study: Gaussian elimination, row-echelon form
Further Readings: Matrix Decompositions

UNIT 2	PROBABILITY DISTRIBUTION	08 HOURS

Application/Case Study: Covid data ananlysis
Contents: Fundamentals of Probability, Random variables and their distributions: Random variables (discrete and continuous), Probability functions, Density and distribution functions, Mean and variance, two dimensional random variables joint probability mass functions, conditioning and independence, Bayes' Theorem, Maximum Likelihood Estimation. Fundamental Sampling Distributions: Random Sampling, Sampling Distributions, Sampling Distribution of Means and the Central Limit Theorem, Sampling Distribution of S2, t–Distribution, F- Distribution.
Self Study: Chi-Square Distribution
Further Readings: Multivariate Central Limit Theorem.

UNIT 3	HYPOTHESIS TESTING	07 HOURS

Application/Case Study: Pharmaceutical Drug Trial

Contents: Parameter and statistic, Concept of null and alternative hypotheses, Acceptance and critical regions, Probabilities of Type I and Type II errors, Level and Power of a test. Concept of p-value, Large sample tests (tests for single mean, difference of means, single proportion, difference of proportions), Tests for small samples (t-test for single mean and difference of means, test for comparison of variances) **Self Study:** Multinomial probability distribution

Further Readings: Factors Affecting power of a Test

UNIT 4	STOCHASTIC PROCESSES AND MARKOV CHAINS	05 HOURS

Application/Case Study: Weather Forecasting

Contents: Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, nstep transition probabilities, Markov chain, Steady state condition, Markov analysis.

Self Study: Irreducible Markov chain

Further Readings: Steady-State (Stationary) Distributions in Markov Chains

UNIT 5	CORRELATION AND REGRESSION ANALYSIS	06 HOURS

Application/Case Study: Analyzing the Relationship Between Height and Weight of Students

Contents: Introduction, Types of Correlation, Karl Pearson Coefficient Correlation, Spearman's Rank Order Correlation, Partial Correlation, Residuals and Plots, Simple Linear Regression, Multiple Regression Model, Least Squares and the Fitted Model, Properties of the Least Squares Estimators, Inferences Concerning the Regression Coefficients, Prediction, Simple Linear Regression Case Study Time-dependent regression: Introduction of Time series Analysis, ARIMA, and SARIMA techniques.

Self Study: Partial Correlation

Further Readings: Probability interpretation of regression and correlation

UNIT 6	NUMERICAL METHODS and CALCULUS APPLICATIONS	07 HOURS

Application/Case Study: Temperature Prediction using Interpolation and Numerical Differentiation

Contents: Numerical Differentiation and Integration, Interpolation: Finite Differences, Newton's and Lagrange's Interpolation. Numerical solution of System of linear equations by Gauss elimination method, LU- Decomposition method.Taylor series, maxima and minima, optimization involving a single variable.

Self Study: Spline Interpolation.

Further Readings: Cubic Splines

TEXT BOOKS

1. Erwin Kreyszig, (2015), Advanced Engineering Mathematics, 10 th edition, Wiley Eastern Ltd.
2. Dr. B.V. Ramana, (2017), Higher Engineering Mathematics, 5 th edition, Tata McGraw Hill.
3. Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, Keying Ye, Probability Statistics for Engineers and Scientists, 9th Ed. Pearson Publishers
4. S. D. Sharma, Operations Research, Kedarnath and Ramnath Publishers, Meerut, Delhi

REFERENCE BOOKS

1. S C Gupta and V K Kapoor, Fundamentals of Mathematical statistics, Khanna publications
2. B.S. Grewal, (2018), Higher Engineering Mathematics, 44 th edition, Khanna Publications.
3. T.T. Soong, Fundamentals of Probability And Statistics For Engineers, John Wiley Sons Ltd, 2004.
4. Sheldon M Ross, Probability and statistics for Engineers and scientists, Academic Press.

E-RESOURCES

1. Linear Algebra By Prof. Dilip P. Patil — IISc Bangalore
2. Introduction to Abstract and Linear Algebra By Prof. Sourav Mukhopadhyay — IIT Kharagpur
3. Numerical Methods for Engineers By Prof. Niket Kaisare — IIT Madras
4. Numerical Methods By Mansoor P — MES College of Engineering Kuttippuram, Kerala
5. Probability and Statistics by By Prof. Somesh Kumar, IIT Kharagpur
6. Introduction to Probability Theory and Stochastic Processes by Dr. S. Dharmaraja, IIT Delhi